

PAC Verification of STL for Linear Parabolic PDEs

Invited Extended Abstract

Jiyu Zhu 

Key Lab. of System Software (Chinese Academy of Sciences)
Institute of Software, Chinese Academy of Sciences
University of Chinese Academy of Sciences
Beijing, China
zhuji@ios.ac.cn

Jie An 

National Key Lab. of Space Integrated Information Systems
Institute of Software, Chinese Academy of Sciences
University of Chinese Academy of Sciences
Beijing, China
anjie@iscas.ac.cn

Shenghua Feng 

National Key Lab. of Space Integrated Information System
Institute of Software, Chinese Academy of Sciences
University of Chinese Academy of Sciences
Beijing, China
fengshenghua@iscas.ac.cn

Naijun Zhan 

School of Computer Science &
MoE Key Lab. of High Confidence Software Technologies
Peking University
Zhongguancun Laboratory, Beijing, China
njzhan@pku.edu.cn

Abstract—This work addresses the PDE verification challenge by introducing a framework for probably approximately correct (PAC) style verification of signal temporal logic (STL) properties over bounded time horizons for linear parabolic PDEs that admit the Feynman–Kac representation. Given the PDE system and a spatial point, with a PAC-style confidence guarantee, the framework can decide whether the trajectory evolving from that point satisfies a given STL specification.

Index Terms—Partial differential equations, Feynman-Kac formula, signal temporal logic, PAC-style verification

I. SYNOPSIS OF OUR APPROACH

Many real-world dynamical processes depend jointly on space and time, and are therefore modeled by partial differential equations (PDEs). Verifying the properties of such systems is more difficult than for ordinary differential equations (ODEs). Techniques that are effective for ODEs often become inapplicable to PDEs. Therefore, unlike ODEs, the verification of PDEs is not well investigated in the literature.

In this work, we focus on a specific but practically meaningful class of PDEs, which admit the *Feynman–Kac representation* [1], and present a probably approximately correct (PAC) verification framework of signal temporal logic (STL) specifications [2] over bounded time horizons.

a) Problem Formalization: We consider linear parabolic PDEs on $\Omega \times [0, T]$ admitting the Feynman–Kac representation:

$$\begin{cases} \frac{\partial u(x, t)}{\partial t} = Au(x, t) - V(x, t)u(x, t) + f(x, t) \\ u(x, t) = u_b(x, t), \quad \forall x \in \partial\Omega, t \in [0, T] \\ u(x, 0) = u_0(x), \quad \forall x \in \Omega \end{cases} \quad (1)$$

where A is the infinitesimal generator given by $Au := \sum_{i=1}^d \mu_i(x, t) \frac{\partial u}{\partial x_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d (\sigma(x, t) \sigma(x, t)^T)_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j}$, V and σ are prescribed functions, $\Omega \subset \mathbb{R}^d$ denotes the compact domain, and $u : \Omega \times [0, T] \rightarrow \mathbb{R}$ denotes the unknown solution.

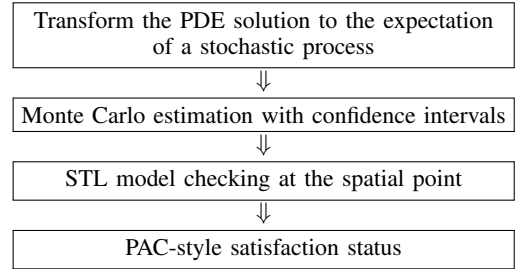


Fig. 1. PAC-style verification framework of STL for Problem (2)

Given the PDE system (1), an STL specification φ , and a monitoring region $X \subset \Omega$, we say the PDE system satisfies φ over X if $\forall x \in X, u(x, \cdot) \models \varphi$. However, directly verifying specifications for PDE systems is challenging. We instead formulate a probably approximately correct (PAC) verification problem: for any $x \in X$, decide whether

$$\Pr(u(x, \cdot) \models \varphi) \geq 1 - \delta \quad (2)$$

where $1 - \delta$ is a prescribed confidence level, and the probability is taken over the randomness introduced by the sampling procedure in the PAC verification framework and represents the confidence level. We say the PDE system (1) *PAC-style* satisfies φ at x if (2) holds.

b) PAC Verification Framework: To address the problem, we propose an PAC verification framework shown in Fig. 1.

We first introduce a Monte Carlo procedure for point-wise value estimation. Assuming $x \in X$ is the spatial point to be checked, the value of a PDE solution $u(x, t)$ at a given time t can be expressed as an expectation over a stochastic path X_s initiated at x , with the contribution of each trajectory determined by whether and when it exits the spatial domain.

The stochastic process X_s is determined by SDE (3):

$$\begin{aligned} dX_s &= \mu(X_s, t-s)ds + \sigma(X_s, t-s)dW_s \\ \text{s.t. } X_0 &= x \in X, s \in [0, t] \end{aligned} \quad (3)$$

where W_s denotes a standard Brownian motion, and the first exit time of X_s from the spatial domain Ω is denoted by $\tau := \inf\{s \geq 0 : X_s \notin \Omega\}$.

We set the starting position of the stochastic path to be x , and then simulate X_s over time. If the stochastic path exits the spatial domain before the time horizon t , then the final-step contribution depends on the boundary condition in (1), i.e., $e^{-\int_0^\tau V(X_\theta, t-\theta)d\theta} u_b(X_\tau, t-\tau)$. Otherwise, the stochastic path remains in the spatial domain, so the final-step contribution depends on the initial condition in (1), i.e., $e^{-\int_0^t V(X_\theta, t-\theta)d\theta} u_0(X_t)$. Taking intermediate contribution $\int_0^{\tau \wedge t} e^{-\int_0^s V(X_\theta, t-\theta)d\theta} f(X_s, t-s)ds$ into account, the complete result is

$$\begin{aligned} u(x, t) &= \mathbb{E}_x \left[\mathbf{1}_{\{\tau > t\}} \cdot e^{-\int_0^t V(X_\theta, t-\theta)d\theta} u_0(X_t) \right. \\ &\quad + \mathbf{1}_{\{\tau \leq t\}} \cdot e^{-\int_0^\tau V(X_\theta, t-\theta)d\theta} u_b(X_\tau, t-\tau) \\ &\quad \left. + \int_0^{\tau \wedge t} e^{-\int_0^s V(X_\theta, t-\theta)d\theta} f(X_s, t-s)ds \right] \end{aligned} \quad (4)$$

We assume that the STL specification is interpreted over the time horizon $[a, b] \subset [0, T]$, which is discretized into m equally spaced time points. We will estimate the PDE solution on those time points. To approximate (4), we discretize the stochastic process using the Euler–Maruyama method and evaluate the associated expectation recursively.

We then employ Hoeffding’s inequality [3] and analyze discretization error of (4) to obtain a Monte Carlo confidence interval $CI(x, t)$, with confidence level $1 - \frac{\delta}{m}$. This means the probability of true solution $u(x, t) \in CI(x, t)$ is $1 - \frac{\delta}{m}$.

We mainly verify the STL specification on a discrete time grid where we apply the above Feynman–Kac representation and Monte Carlo simulation to obtain values on discrete points. Under mild assumptions in parabolic Schauder theory [4], the PDE solution to (1) is Lipschitz continuous with respect to time. STL verification against a discrete trajectory can be reduced to an offline monitoring problem. Given an STL formula with the “global” operator $G_{[a,b]}(u(x, \cdot) \bowtie h)$ where $\bowtie \in \{=, <, \leq, >, \geq\}$, for each discrete time point t_j within $[a, b]$, if for all $r \in CI(x, t_j)$ satisfies $r \bowtie h$, then we say the PDE system (1) PAC-style satisfies φ at x . Conversely, we say (1) does not PAC-style satisfy φ at x . Addressing a formula with the “eventually” operator $F_{[a,b]}\varphi$, we say (1) PAC-style satisfies φ at x if there exists $t_j \in [a, b]$ such that $(u(x, \cdot), t_j) \models \varphi$. For a formula with the “until” operator $\varphi_1 U_{[a,b]} \varphi_2$, we search for a witness time t_k such that $(u(x, \cdot), t_k) \models \varphi_2$ and $(u(x, \cdot), t_l) \models \varphi_1$ are true on every intermediate time steps $l = j, \dots, k-1$. If such a witness time point exists, we say (1) PAC-style satisfies φ at x .

Since for each discrete time point, we construct the confidence interval with failure probability $\frac{\delta}{m}$, by the union bound, the probability that all m discrete points lie within their corresponding confidence intervals is at least $1 - \delta$. If the

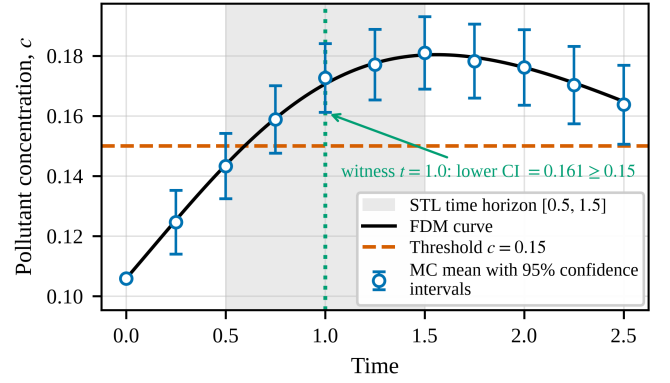


Fig. 2. Pollutant concentration at the spatial point $x = (2.4, 2)$: FDM curve and Monte Carlo means and 95% confidence intervals.

monitoring algorithm returns a conclusive satisfaction result on the discrete trajectory, we can say that $u(x, \cdot) \models \varphi$ at spatial point x with probability $1 - \delta$.

c) *Preliminary Experiments*: Consider a 2-D pollutant transport problem $c_t + v \cdot \nabla c = \nabla \cdot (D \nabla c) - kc + f$ where $c(x, t)$ denotes the pollutant concentration, $v \cdot \nabla c$ represents the pollutant transport with flow, $\nabla \cdot (D \nabla c)$ represents diffusion, $-kc$ represents pollutant removal and f represents a local pollutant source. The goal is to determine if the system satisfies STL formula $\varphi = F_{[0.5, 1.5]}(c \geq 0.15)$ at point $x = (2.4, 2)$ with confidence 0.95. To compare with numerical methods, e.g., the finite difference method (FDM), we let discrete time points be $0, 0.25, \dots, 2.5$ with 5,000 Monte Carlo sample paths per time point. We observe that at $t = 1.0$, the 95% confidence interval is $[0.161, 0.184]$, whose lower bound is larger than 0.15, thus we can say that this system PAC-style satisfies φ at $x = (2.4, 2)$ with confidence 0.95. The FDM curve in Fig. 2 is obtained by solving the PDE over the full space-time domain and extracting the solution at $x = (2.4, 2)$.

II. CONCLUSION AND FUTURE WORK

This extended abstract presents a novel preliminary framework for PAC-style verification of STL over bounded-time trajectories for a class of linear parabolic PDEs. Note that, for the detailed techniques in our approach, both simulating the stochastic paths and querying the satisfaction status of the spatial points are naturally suitable for parallel computing. We plan to thoroughly analyze the discretization error of the Monte Carlo simulation and reduce the conservativeness of our framework by applying tighter estimation of confidence bounds and interval-monitoring procedures.

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